



PT SAPTAINDRA SEJATI

TORQUEFLOW

RECEIVING INSPECTION

CHECKSHEET

MACHINE MODEL

HD785-7

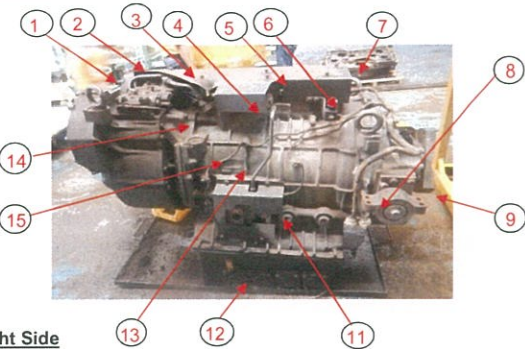
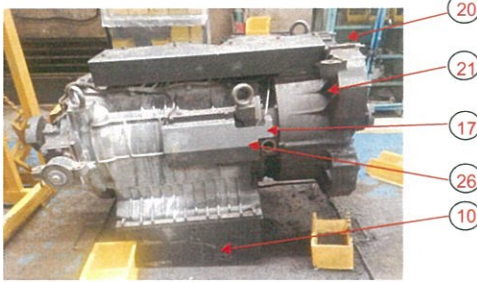
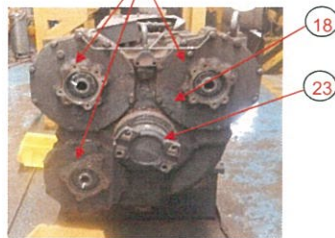
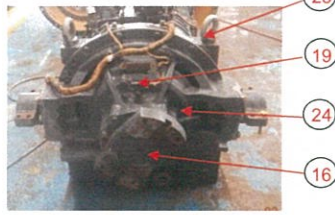
RO NO :

POWERTRAIN SECTION

RECEIVING INSPECTION SHEET TORQUEFLOW	MACHINE MODEL	HD785-7	PART NO	561-15-71001	HOUR METER	
	DOCUMENT NO	RC/QR01/PT/05/02	COMP. S/N		LIFE TIME	
	RECEIVING DATE		SITE		REVISION NO	
	HISTORICAL		WO NO		PAGE	

BELOW ARE THE ITEMS RELATED TO DELIVERY INSPECTION AND WHILE YOU DOING THIS INSPECTION JOB, PLEASE REFERRING TO THE REMANUFACTURED COMPONENTS - STANDARD CONFIGURATION.

✓ GOOD CONDITION X : BAD CONDITION ⊗ : NONE

SKETCH DRAWING OR PHOTOES	NO.	ITEM	INSPECTION CONDITION	CHECK	
				IN	OUT
Left Side 	1	Torque Converter			
		Model			
		Serial Number			
		WO.no			
Right Side 	2	T/C Valve			
	3	Lock Up Valve			
	4	Transmission			
		Model			
Front view 		Serial Number			
		WO.no			
	5	Oil Filter ECMV			
	6	Breather			
Rear view 	7	ECMV Cover			
	8	Connector & Wiring Harness			
	9	Rear Bracket LH,RH			
	10	Drain Valve Oil			
	11	Lubricating Valve			
	12	Oil Pan			
	13	T/M Housing			
	14	Bolt Connection			
	15	Temperature Sensor			
	16	T/M name plate			
	17	Intermediate Shaft Speed Sensor			
	18	Front bracket			
	19	Out put shaft Speed Sensor			
	20	T/C name plate			
	21	T/C Housing			
	22	PTO Spline			
	23	Front Coupling			
	24	Rear coupling			
	25	Eye Bolt			
	26	Cover			
		Stand (Original/STD)			
		Painting			
		Packing			
NOTE PLEASE MARK " " FOLLOWING TO THE ACTUAL CONDITION					
OIL LEVEL	EMPTY OR DRAINED	✓			
	PROPER LEVEL				

Jakarta,

Inspected & Prepared by

Check by

Check by

Approved by



PT SAPTAINDRA SEJATI

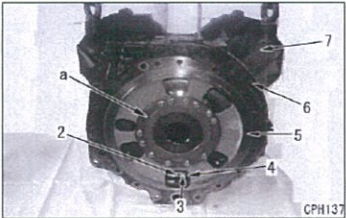
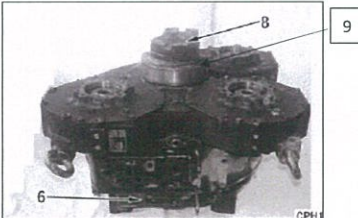
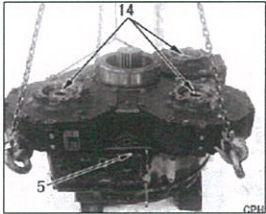
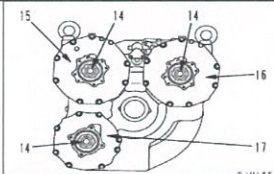
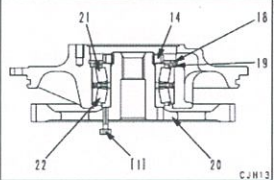
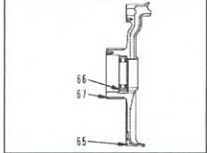
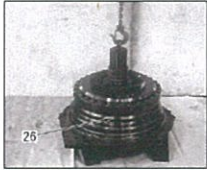
**PTO & TORQUE CONVERTER
DISASSEMBLY
CHECKSHEET**

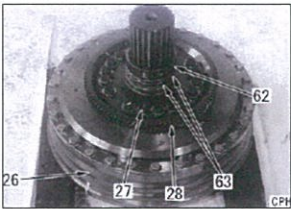
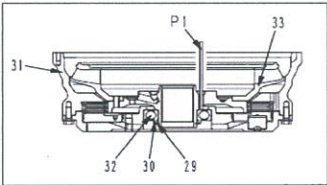
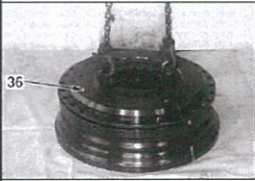
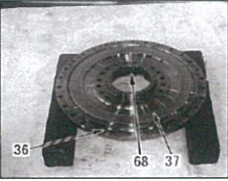
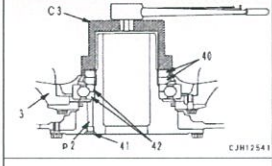
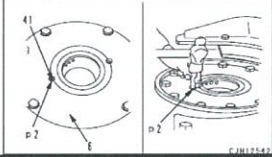
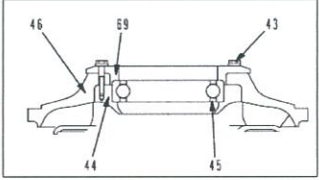
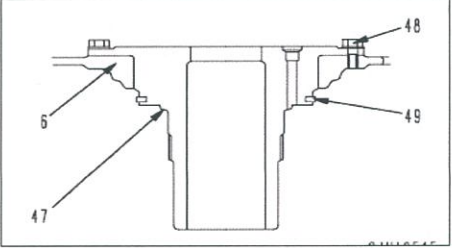
MACHINE MODEL

HD785-7

RO NO :

POWERTRAIN SECTION

DISASSEMBLY CHECK SHEET		WO NUMBER	DOCUMENT NO
TORQUE CONVERTER & PTO		COMP.MODEL	PUBLIC DATE
HD 785 - 7		COMP.S/N	REVISION NO
		UNIT MODEL	PAGE
PROCESS	START FINISH	DISASSEMBLY BY	
NO	PART TO REMOVE OR INSPECT	CONDITIONS TO BE INSPECTION	MECH GL SKEETH DRAWING OR PHOTOES REMARK
1	PTO & TORQUE CONVERTER ASSEMBLY	- Remove the attached parts and relief valve assembly (1) - Remove drain plug to drain oil. - Remove the mounting bolts (4) of the pump assembly (3). - Remove the PTO assembly mounting bolts (7) by leaving 4 bolts evenly spaced. - Remove orifice plug (a) so that it will not be broken.	
2	TORQUE CONVERTER ASSY	Set the assembly with torque converter housing assembly (6) facing downward. - Remove input coupling (8). - Pull out cage assembly (9) by using forcing screws. - Disassembly of cage assembly - Remove O-ring (10). - Remove oil seal (11). - Remove dust seal (12). - Remove snap ring (13). - Lock the pump drive gear. Loosen nuts (14) using tool C3. - Disconnect the PTO assembly (5) from the rear cover assembly	  
	PTO ASSEMBLY	- Remove the pump drive gear cage assemblies (15), (16) and (17) - Disassembly of pump drive gear cage assembly - Remove nut (14). Remove snap ring (18) and remove plate (19). - Use a forcing screw [1] to remove gear (20). - Remove bearings (21) and (22). - Use push tool to remove outer races from cover	 
4	PTO CASE	- Remove bushing (67) and outer race (66) from PTO case (65). - Remove turbine clutch assembly (26).	 

DISASSEMBLY CHECK SHEET TORQUE CONVERTER & PTO HD 785 - 7		WO NUMBER			DOCUMENT NO	
		COMP.MODEL			PUBLIC DATE	
		COMP.S/N			REVISION NO	
		UNIT MODEL			PAGE	
PROCESS		START			DISASSEMBLY BY	
		FINISH				
NO	PART TO REMOVE OR INSPECT	CONDITIONS TO BE INSPECTION	MECH	GL	SKETCH DRAWING OR PHOTOES	REMARK
5	TURBINE CLUTCH ASSEMBLY	- Drive gear assembly Remove gear mounting bolts (27). Remove the drive gear assembly (28) using the forcing screw. Remove seal ring (63). Remove inner race (62). - Remove snap ring (29) and remove spacer (30). - Reverse turbine clutch assembly (31). Hammer out the bearing inner race from the holes on the boss to remove turbine assembly (33). - Remove inner gear mounting bolts (35) from the turbine (34).			 	
6	TURBINE CLUTCH ASSEMBLY	Remove the housing assembly (36) with the housing facing upward. Remove piston (37) and bearing (68) from the housing (36). Remove disc (38) and plate (39).			 	
7	STATOR ASSY	Turn stator assembly (57) clockwise, pull up and remove.				
8	PUMP ASSEMBLY	Use tool C3 to remove nut (40). Reverse the pump and torque converter housing assembly and place the assembly down. Remove plug (41). Hammer inner race Sling torque converter housing (6) to remove pump assembly (3).			 	
		Remove retainer (69) mounting bolts Remove retainer (69). Remove guide (44). Remove bearing (45) from pump (46).				
9	SHAFT ASSEMBLY	Remove shaft (47) and mounting bolts Reverse and install the torque converter housing assembly (6). Hammer and remove shaft (47). Remove seal ring (49). a This procedure is not required if the shaft is not to be replaced.				

DISASSEMBLY CHECK SHEET TORQUE CONVERTER & PTO HD 785 - 7		WO NUMBER COMP.MODEL COMP.S/N UNIT MODEL	DOCUMENT NO PUBLIC DATE REVISION NO PAGE
PROCESS	START FINISH		DISASSEMBLY BY

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PT SAPTAINDRA SEJATI

TRANSMISSION

DISASSEMBLY

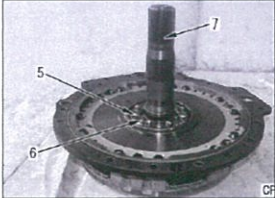
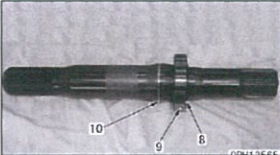
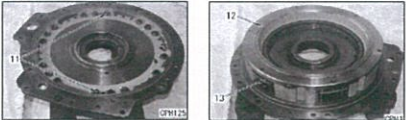
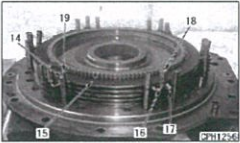
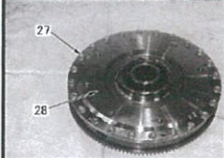
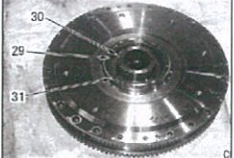
CHECKSHEET

MACHINE MODEL

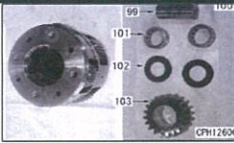
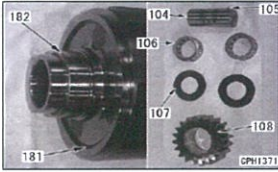
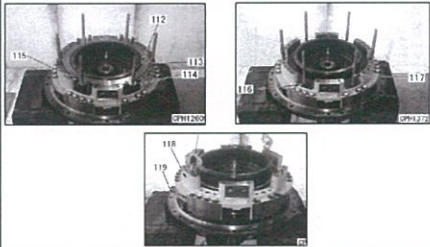
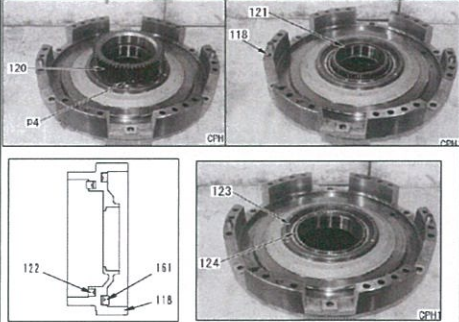
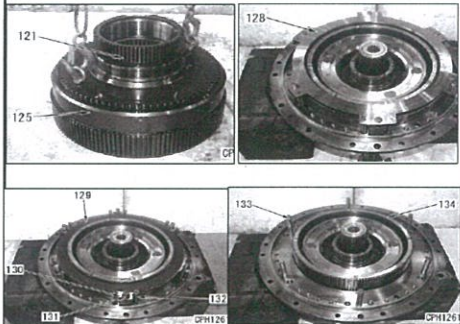
HD785-7

RO NO :

POWERTRAIN SECTION

DISASSEMBLY CHECK SHEET		WO NUMBER	DOCUMENT NO		RC/QR02/PT/05/02	
TRANSMISSION		COMP.MODEL	PUBLIC DATE			
HD 785 - 7		COMP.S/N	REVISION NO			
		UNIT MODEL	PAGE			
PROCESS	START	DISASSEMBLY BY				
	FINISH					
NO	PART TO REMOVE OR INSPECT	CONDITIONS TO BE INSPECTION	MECH	GL	SKETCH DRAWING OR PHOTOS	REMARK
	CLUTCH NO 1	When disassembling the clutch disc and plate, group the parts by each Number. and put tags to identify them Remove 5 sleeves Remove clutch No. 1 assembly Input shaft Remove bolts and plate from shaft assembly . Sling and remove input shaft assembly Remove snap ring (8) from the input shaft and remove bearing (9). Remove seal ring (10). Housing and piston Remove the remaining 2 bolts and remove housing and piston assembly (12). Remove piston (13) and seal ring from the housing. Remove piston (13) and seal ring from the housing Guide spring, disc, spring, plate and guide pin Remove spring (14) and spring guide pin (19). Remove disc (15), spring (16), and plate (17). Remove guide pin (18). No. 1 clutch and ring gear Remove ring gear (21) and dowel pins (22) and (23) from cage (20). a Remove the snap ring (183) from ring gear (21). Bearing Remove snap ring (24) and then remove bearing (25) from cage (20).			 CLUTCH NO 1  INPUT SHAFT   	
2	CLUTCH NO 2	Housing assembly Remove mounting bolts (27) to remove housing assembly (28). Remove snap ring (29) and remove bushing (30). Remove inner race (31). Remove mounting bolts (32) to remove cylinder (33) from housing assembly (34). Remove piston (35) from housing Remove seal ring (36) from piston Remove seal rings and from the housing (34). Remove bearing (39) from the housing 2) Disc, plate and spring Remove disc, plate, and spring 3) Transmission input shaft rotation detecting plate 1) Reverse and install the clutch assembly. 2) Remove mounting bolts (43) to remove plate (44).			     	

DISASSEMBLY CHECK SHEET TRANSMISSION HD 785 - 7		WO NUMBER	DOCUMENT NO	RC/QR02/PT/05/02		
		COMP.MODEL	PUBLIC DATE			
		COMP.S/N	REVISION NO			
		UNIT MODEL	PAGE			
PROCESS		START	DISASSEMBLY BY			
		FINISH				
NO	PART TO REMOVE OR INSPECT	CONDITIONS TO BE INSPECTION	MECH	GL	SKETCH DRAWING OR PHOTOES	REMARK
		4) Planetary gear 1) Remove carrier assembly (46) from drum (45). 2) Remove shaft (47). a Care should be taken not to lose ball (48). 3) Remove gear (49), bearing (50) and thrust washer (51). 4) Remove bearing (52). 5) Remove inner race (177) and seal ring (178) from carrier assembly (46).				
3	TRANSMISSION CASE	Remove the following parts from transmission case (53). 1) Cage (54) 2) Bearing (55) 3) Collar (56) 3) Remove seal ring (57) from collar (56).				
4	CLUTCH HOUSING NO 3 ASSEMBLY	For disassembly of the housing assembly, refer to the "Disassembly of clutch No. 2 assembly." Ring gear No. 3 1) Remove ring gear No. 3 (59). 2) Remove inner race (60) from ring gear (59). 3) Remove seal ring (61) from ring gear (59). Remove tie bolts (62).				
5	Clutch No. 4 disc, plate and spring	Remove spring (64). Remove discs (65), plates (66) and springs (67). Remove plate (68). Remove snap ring (69). Remove sun gear (70).				
6	Housing No. 4, piston and ring gear&Clutch No. 5 disc, plate and spring	Housing No. 4, piston and ring gear remove housing No. 4 (71), piston (72) and ring gear (73) in one assembly. Clutch No. 5 disc, plate and spring Remove disc, plate and springs (76) and (77). Remove plate				
7	Carrier No. 4 assembly& ring gear No. 5	Remove carrier No. 4, ring gear No. 5 assembly Disconnect carrier No. 4 assembly (80) and ring gear No. 5 (81).				
8	Planetary gear No. 4	Pull out shaft (83) toward the side of the ball from carrier (80). Remove ball (84) from carrier (80). Remove gear (85), bearing (86) and thrust washer (87) from carrier (80). Remove bearing (89) from carrier(88) remove carrier No. 5 and No. 6 assembly (90).				
9		Remove collar (180) and then remove the snap ring and inner race from it. Remove bolts (91) from carrier assembly (90) and remove cover (92). Sling and remove the shaft and sun gear assembly. Pull out sun gear from shaft.				
		Shaft Remove snap ring from the shaft. Remove inner race from the shaft.				

DISASSEMBLY CHECK SHEET TRANSMISSION HD 785 - 7		WO NUMBER	DOCUMENT NO		RC/QR02/PT/05/02	
		COMP.MODEL	PUBLIC DATE			
		COMP.S/N	REVISION NO			
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PROCESS		START FINISH	DISASSEMBLY BY			
NO	PART TO REMOVE OR INSPECT	CONDITIONS TO BE INSPECTION	MECH	GL	SKETCH DRAWING OR PHOTOS	REMARK
10	CARRIER No. 5 ASSEMBLY	Disassembly of carrier No. 5 assembly " Remove shaft (99), ball (100), bearing (101), thrust washer (102) and gear (103) accordingly.				
11	CARRIER No. 6 ASSEMBLY	Remove snap ring (181) from the No. 6 carrier assembly. Remove shaft (104), ball (105), bearing (106), thrust washer (107) and gear (108) accordingly. Remove inner race (182) from the No.6 carrier assembly				
12	HOUSING NO 5, PISTON AND SPRING GUIDE PIN	Remove spring guide pin (111). remove housing No. 5 (109) and piston (110) in one assembly.				
13	CLUTCH DISC NO 6, PLATE AND SPRING	Remove disc (112), plate (113) and springs (114) and (115) Remove plate (116). Remove guide pin (117). Remove housing Nos. 6, 7 assembly (118) using the eyebolts.				
14	RING GEAR NO 6	Housing Raise and remove the sun gear by pushing the snap ring inside through three holes on the circumference of sun gear (120). Hammer and let gear No. 7 (121) fall down to disconnect Sling and remove housing assembly (118). Remove piston No. 6 and No. 7 from housing. Remove snap ring (123) & Remove bearing (124).				
15	CLUTCH NO 7	Gear No. 7 Disconnect ring gear No. 6 (125) and gear No. 7 (121). Remove plate (128). Remove discs (129), plates (130) and springs (131) alternately. Remove guide springs (132). Remove guide pin Remove clutch ring gear No. 7				

DISASSEMBLY CHECK SHEET TRANSMISSION HD 785 - 7		WO NUMBER	DOCUMENT NO	RC/QR02/PT/05/02		
		COMP.MODEL	PUBLIC DATE			
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		UNIT MODEL	PAGE			
PROCESS	START FINISH		DISASSEMBLY BY			
NO	PART TO REMOVE OR INSPECT	CONDITIONS TO BE INSPECTION	MECH	GL	SKETCH DRAWING OR PHOTOES	REMARK
16	OUTPUT SHAFT ASSEMBLY	Remove output shaft and cage assembly (136) from cover assembly (119). Remove collar from the output shaft. Receive cage (138), push output shaft (139) with a press to pull it out from the bearing. Remove bearing from output shaft Remove bearing (141) from cage (138). Remove oil seal (142) from cage (138). Remove output shaft rotation sensor gear Remove O-ring (148). 3) Remove dowel pin (144). 4) Remove snap ring (145) and spacer (146). 5) Hammer carrier assembly (147) to let it fall to remove. a Place a piece of soft cloth and so forth under the carrier assembly. 6) Remove bearing (160).				
17	CARRIER NO 7	Remove mounting bolts to remove plate. Remove seal ring (151) from carrier (147). Remove shaft (152), ball (153), bearing (154), thrust washer (155) and gear (156) from carrier (147)				
18	BEARING	Reverse cover assembly (119). Remove cage mounting bolts (157) to Remove cage (158). Remove seal ring (159). Remove bearing (160) from cover (119).				

NOTE :

INSPECTION BY

APPROVED BY

MECHANIC

GL /QA



PT SAPTAINDRA SEJATI

TORQUEFLOW

DISASSEMBLY & MEASUREMENT

CHECKSHEET

MACHINE MODEL

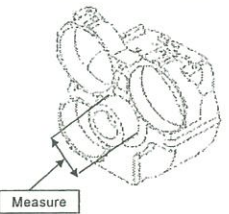
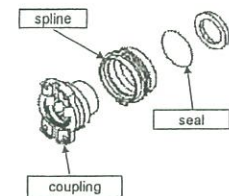
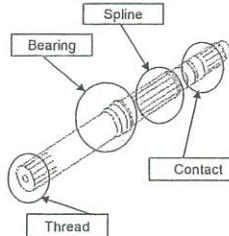
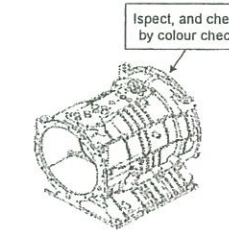
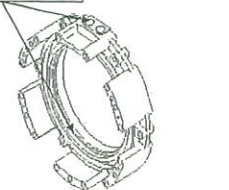
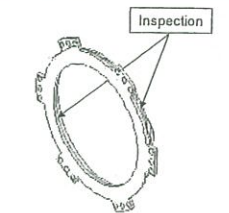
HD785-7

RO NO :

POWERTRAIN SECTION

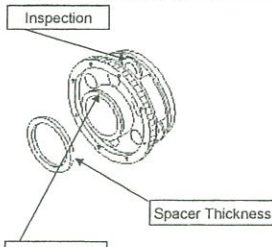
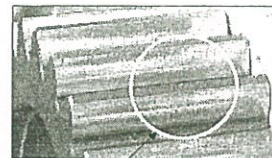
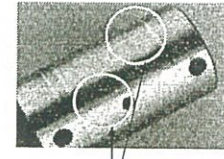
DISASSEMBLY AND MEASUREMENT CHECK SHEET TORQUEFLOW HD 785-7	UNIT MODEL	DOCUMENT NO	
	COMP. MODEL	PUBLIC DATE	15/10/2009
	COMP. S/N	REVISION NO.	00
	WO.NO	PAGE	1 of 6

PROCESS	START FINISH	DISASSEMBLY BY
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No	PARTS NAME	METHODE	CHECK POINT	EVALUATION ACTUAL	DECISION U/A U/R R/N	FIGURE OF PARTS TO BE CHECKED																																														
1	SUPPORT	MEASUREMENT	Dimension at contact area with bracket (Trunion) Compare with standard size <table><tr><th>UNIT MODEL</th><th>STD SIZE</th></tr><tr><td>HD465-5</td><td>229,975 ~ 229,98 mm</td></tr><tr><td>HD 785-7</td><td>229,975 ~ 229,98 mm</td></tr><tr><td>D275A-2</td><td>234,995 ~ 235,00 mm</td></tr><tr><td>D375A-2 / 3</td><td>234,995 ~ 235,00 mm</td></tr><tr><td>D475A-2 / 3</td><td>234,995 ~ 235,00 mm</td></tr></table>	UNIT MODEL	STD SIZE	HD465-5	229,975 ~ 229,98 mm	HD 785-7	229,975 ~ 229,98 mm	D275A-2	234,995 ~ 235,00 mm	D375A-2 / 3	234,995 ~ 235,00 mm	D475A-2 / 3	234,995 ~ 235,00 mm	<table><tr><th>MODEL</th><th>DIMENSION ACT</th></tr><tr><td></td><td></td></tr></table>	MODEL	DIMENSION ACT			<table><tr><td></td><td></td><td></td></tr></table>																															
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D475A-2 / 3	234,995 ~ 235,00 mm																																																			
MODEL	DIMENSION ACT																																																			
2	COUPLING (FLANGE)	MEASUREMENT VISUAL CHECK	Worn at spline and seal contact area Worn at surface of contact u-joint Compare with standard size Seal contact size : <table><tr><td>DEPTH : STD</td><td>≤ 0.1 mm</td></tr><tr><td>WIDTH : STD</td><td>≤ 0.5 mm</td></tr></table> <i>Spline wear standard ≤ 0,35 mm</i> Mating face (u-joint) contact <i>Depth ≤ 0,2 mm</i>	DEPTH : STD	≤ 0.1 mm	WIDTH : STD	≤ 0.5 mm	<table><tr><th>SEAL CONTACT</th></tr><tr><th>DEPTH</th></tr><tr><td></td><td></td></tr></table> <table><tr><th>SPLINE WEAR</th></tr><tr><td></td></tr></table> <table><tr><th>MATING FACE</th></tr><tr><td></td></tr></table>	SEAL CONTACT	DEPTH			SPLINE WEAR		MATING FACE		<table><tr><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td></tr></table>																																			
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3	INPUT SHAFT	MEASUREMENT VISUAL CHECK	Spline teeth wear, Deformation, Crack, Breakage <i>Wear dimension ≤ 0,35 mm.</i> <i>Seal Ring groove wear ≤ 0,2 mm</i> Thread Portion condition Bearing contact area Bearing Seat surface	<table><tr><th>WEAR (%)</th></tr><tr><td></td></tr></table> <table><tr><th>WEAR (%)</th></tr><tr><td></td></tr></table>	WEAR (%)		WEAR (%)		<table><tr><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td></tr></table>																																											
WEAR (%)																																																				
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4	CASE ASS'Y	VISUAL CHECK	Check all of position at case ass'y from Crack, Damage at Thread hole , worn at surface contact of bearing seat Mating Face condition from Scratch, Crack and Damage.	<table><tr><th>CONDITION</th><th>% Targe</th></tr><tr><td>Crack</td><td></td></tr><tr><td>Damage</td><td></td></tr><tr><td>Wear</td><td></td></tr></table>	CONDITION	% Targe	Crack		Damage		Wear		<table><tr><td></td><td></td><td></td></tr></table>																																							
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Wear																																																				
5	HOUSING CLUTCH	VISUAL CHECK MEASUREMENT	Wear and damage of seal ring contact surface Wear and damage of inside and outside surface of piston insertion portion Wear and damage of ring gear sliding surface <i>wear std less than 0,3 mm</i>	<table><tr><th>CONDITION</th><th>CLUTCH</th></tr><tr><td rowspan="3">Seal Contact</td><td>Crack</td></tr><tr><td>Damage</td></tr><tr><td>Wear</td></tr><tr><td rowspan="3">Ring gear Contact</td><td>Wear</td></tr><tr><td>Damage</td></tr><tr><td></td></tr><tr><td rowspan="3">Piston seal</td><td>Crack</td></tr><tr><td>Damage</td></tr><tr><td>Wear</td></tr></table> <table><tr><th>H</th></tr><tr><td>L</td></tr><tr><td>R</td></tr><tr><td>F</td></tr><tr><td>4</td></tr><tr><td>3</td></tr><tr><td>2</td></tr><tr><td>1</td></tr></table>	CONDITION	CLUTCH	Seal Contact	Crack	Damage	Wear	Ring gear Contact	Wear	Damage		Piston seal	Crack	Damage	Wear	H	L	R	F	4	3	2	1	<table><tr><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td></tr></table>																									
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6	PISTON	VISUAL CHECK MEASUREMENT	Wear and damage of seal ring contact surface Wear and damage of inside and outside surface of piston Wear and damage of Disc contact surface <i>Wear std less than 0,2 mm</i>	<table><tr><th>CONDITION</th><th>CLUTCH</th></tr><tr><td rowspan="3">Seal Contact</td><td>Crack</td></tr><tr><td>Damage</td></tr><tr><td>Wear</td></tr><tr><td rowspan="3">Disc Contact</td><td>Wear</td></tr><tr><td>Damage</td></tr><tr><td></td></tr><tr><td rowspan="3">Hole pin</td><td>Crack</td></tr><tr><td>Damage</td></tr><tr><td>Wear</td></tr></table> <table><tr><th>H</th></tr><tr><td>L</td></tr><tr><td>R</td></tr><tr><td>F</td></tr><tr><td>4</td></tr><tr><td>3</td></tr><tr><td>2</td></tr><tr><td>1</td></tr></table>	CONDITION	CLUTCH	Seal Contact	Crack	Damage	Wear	Disc Contact	Wear	Damage		Hole pin	Crack	Damage	Wear	H	L	R	F	4	3	2	1	<table><tr><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td></tr></table>																									
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DISASSEMBLY AND MEASUREMENT CHECK SHEET	UNIT MODEL		DOCUMENT NO	
	COMP. MODEL		PUBLIC DATE	15/10/2009
	COMP. S/N		REVISION NO.	00
	WO.NO		PAGE	2 of 6

TORQUEFLOW
HD785-7

No	PARTS NAME	METHODE	CHECK POINT	EVALUATION			DECISION			FIGURE OF PARTS TO BE CHECKED
				ACTUAL			U/A	U/R	R/N	
7	RING GEAR	MEASUREMENT VISUAL CHECK	Internal gear teeth surface (Pitting , Crack, Damage Breakage, Abnormal Wear) <i>Max wear less than 0,5 mm</i> External tooth flank in contact with disc (Stepper wear, Crack, Chipping) Housing sliding surface. <i>std max wear less than 0,3 mm</i>	Clutch	POSITION	CONDITION	H			
				H			L			
				L			R			
				R			F			
				F			4			
				4			3			
				3			2			
				2			1			
				1						
8	CARRIER	MEASUREMENT VISUAL CHECK	Shaft Boss (Crack, Distortion,) Seat Bearing Diameter, Spacer Bearing Gear teeth Pitting, Crack, Breakage, Abnormal wear. Planetary Shaft Hole (Fatigue , Wear) <i>Max Wear less than 0,1 mm</i> Wear at surface contact with trust washer <i>Max wear less than 0,2 mm</i>	Clutch	POSITION	CONDITION	H			
				H			L			
				L			R			
				R			F			
				F			4			
				4			3			
				3			2			
				2			1			
				1						
	SPACER	MEASUREMENT	Thickness wear, Inner Diameter, Outer Diameter							
9	SUN GEAR	VISUAL CHECK	Gear Surface, Pitting, Crack, Breakage, Abnormal wear INSPECTION AREA	CONDITION	PITTING	CRACK	SCORING	H		
				CLUTCH				L		
				HIGHT				R		
				LOW				F		
				REVERSE				4		
				FORWARD				3		
				1				2		
				2				1		
				3						
				4						
10	PLANETARY GEAR	VISUAL CHECK	Gear Surface, Pitting, Crack, Breakage, Abnormal wear at inside diameter of gear	CONDITION	PITTING	WEAR	SCORING	H		
				CLUTCH	(PCS)	(PCS)	(PCS)	L		
				HIGHT				R		
				LOW				F		
				REVERSE				4		
				FORWARD				3		
				1				2		
				2				1		
				3						
				4						
				Note : Fill in the blank with Qty of gear damage						
11	SHAFT GEAR	VISUAL CHECK MEASUREMENT	Wear at surface contact with bearing Wear at surface contact with thrust washer Crack (Damage) at hole of oil galary.	CONDITION	CRACK	WEAR	PITTING	H		
				CLUTCH	(PCS)	(PCS)	(PCS)	L		
				HIGHT				R		
				LOW				F		
				REVERSE				4		
				FORWARD				3		
				1				2		
				2				1		
				3						
				4						
				Note : Fill in the blank with Qty of Shaft damage						
12	HOUSING (END PLATE)	VISUAL CHECK MEASUREMENT	Disc contact surface heat spot metal transfer Wear at disc contact surface <i>std max less than 0,2 mm</i> Wear at ring gear contact surface <i>std max less than 0,2 mm</i>	CONDITION	% Tg					
				Heat Spot						
				Damage						
				Wear						



DISASSEMBLY AND MEASUREMENT CHECK SHEET TRANSMISSION FOR ALL MODEL	UNIT MODEL		DOCUMENT NO	
	COMP. MODEL		PUBLIC DATE	15/10/2009
	COMP. S/N		REVISION NO.	00
	WO.NO		PAGE	3 of 6

No	PARTS NAME	MEASUREMENT DATA	FIGURE OF PARTS TO BE CHECKED																																																																																																																																																																																																																																																																																																																																																																																																																																																																												
13	SPRING CLUTCH	VISUAL CHECK MEASUREMENT DIMENSION OF STANDARD FREE LENGTH (mm) <table><tr><th>MODEL CLUTCH</th><th>HD 465-5/7</th><th>HD785-7</th><th>GD825A-2</th><th>WA600</th><th>WA600</th><th>WA800</th><th>D85ES-2</th><th>D155AX-3</th><th>D275A-2/3</th><th>D375A-2/3</th></tr><tr><td>HIGH</td><td>42.00</td><td>52.00</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>LOW</td><td></td><td></td><td>66.00</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>REVERSE</td><td>91.00</td><td>84.00</td><td>53.00</td><td>68.00</td><td>68.00</td><td>91.00</td><td>59.00</td><td>56.80</td><td>59.00</td><td>66.00</td></tr><tr><td>FORWARD</td><td></td><td></td><td>53.00</td><td>68.00</td><td>68.00</td><td>108.00</td><td>69.50</td><td>56.80</td><td>70.00</td><td>70.00</td></tr><tr><td>1</td><td>68.00</td><td>68.00</td><td>39.60</td><td>62.00</td><td>52.00</td><td>91.00</td><td>54.00</td><td>39.80</td><td></td><td></td></tr><tr><td>2</td><td>68.00</td><td>68.00</td><td>39.60</td><td></td><td></td><td>91.00</td><td>54.00</td><td>84.00</td><td>91.00</td><td>97.00</td></tr><tr><td>3</td><td>52.00</td><td>41.50</td><td>39.60</td><td>62.00</td><td>52.00</td><td></td><td>54.00</td><td>56.80</td><td>91.00</td><td>97.00</td></tr><tr><td>4</td><td></td><td></td><td>39.60</td><td>52.00</td><td>52.00</td><td></td><td></td><td></td><td></td><td></td></tr></table> <table><tr><th>SIZE CLUTCH</th><th colspan="12">ACTUAL SIZE OF FREE LENGTH (mm)</th></tr><tr><td></td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td><td>9</td><td>10</td><td>11</td><td>12</td></tr><tr><td>HIGHT</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>LOW</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>REVERSE</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>FORWARD</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>1</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>2</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>3</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>4</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr></table>	MODEL CLUTCH	HD 465-5/7	HD785-7	GD825A-2	WA600	WA600	WA800	D85ES-2	D155AX-3	D275A-2/3	D375A-2/3	HIGH	42.00	52.00									LOW			66.00								REVERSE	91.00	84.00	53.00	68.00	68.00	91.00	59.00	56.80	59.00	66.00	FORWARD			53.00	68.00	68.00	108.00	69.50	56.80	70.00	70.00	1	68.00	68.00	39.60	62.00	52.00	91.00	54.00	39.80			2	68.00	68.00	39.60			91.00	54.00	84.00	91.00	97.00	3	52.00	41.50	39.60	62.00	52.00		54.00	56.80	91.00	97.00	4			39.60	52.00	52.00						SIZE CLUTCH	ACTUAL SIZE OF FREE LENGTH (mm)													1	2	3	4	5	6	7	8	9	10	11	12	HIGHT													LOW													REVERSE													FORWARD													1													2													3													4													Straightness at coil spring, Crack or Discolouration Free length of spring, compare with standard as follow Dimension of spring free length as follow <table><tr><th colspan="4">DECISION</th></tr><tr><td>CLT</td><td>U/A</td><td>U/R</td><td>R/N</td></tr><tr><td>H</td><td></td><td></td><td></td></tr><tr><td>L</td><td></td><td></td><td></td></tr><tr><td>R</td><td></td><td></td><td></td></tr><tr><td>F</td><td></td><td></td><td></td></tr><tr><td>1</td><td></td><td></td><td></td></tr><tr><td>2</td><td></td><td></td><td></td></tr><tr><td>3</td><td></td><td></td><td></td></tr><tr><td>4</td><td></td><td></td><td></td></tr></table>	DECISION				CLT	U/A	U/R	R/N	H				L				R				F				1				2				3				4																																																																																																																																																																																																		
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Plates of 400 mm or more in O.D. Less than 0,6 mm b. Plates of less than 400 mm in O.D Less than 0,3 mm c. Brake Plates deformation less than 0,7 mm. In Dish a. Plates of 400 mm or more in O.D. less less than 0,2 mm b. Plates of less than 400 mm in O.D. less than 0,1 mm. <table border="1"> <thead> <tr> <th colspan="4">DECISION</th></tr> <tr> <th>CLT</th><th>U/A</th><th>U/R</th><th>R/N</th></tr> </thead> <tbody> <tr> <td>H</td><td></td><td></td><td></td></tr> <tr> <td>L</td><td></td><td></td><td></td></tr> <tr> <td>R</td><td></td><td></td><td></td></tr> <tr> <td>F</td><td></td><td></td><td></td></tr> <tr> <td>1</td><td></td><td></td><td></td></tr> <tr> <td>2</td><td></td><td></td><td></td></tr> <tr> <td>3</td><td></td><td></td><td></td></tr> <tr> <td>4</td><td></td><td></td><td></td></tr> </tbody> </table>	DECISION				CLT	U/A	U/R	R/N	H				L				R				F				1				2				3				4			
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PT SAPTAINDRA SEJATI

CONTROL VALVE

DISASSEMBLY

CHECKSHEET

MACHINE MODEL

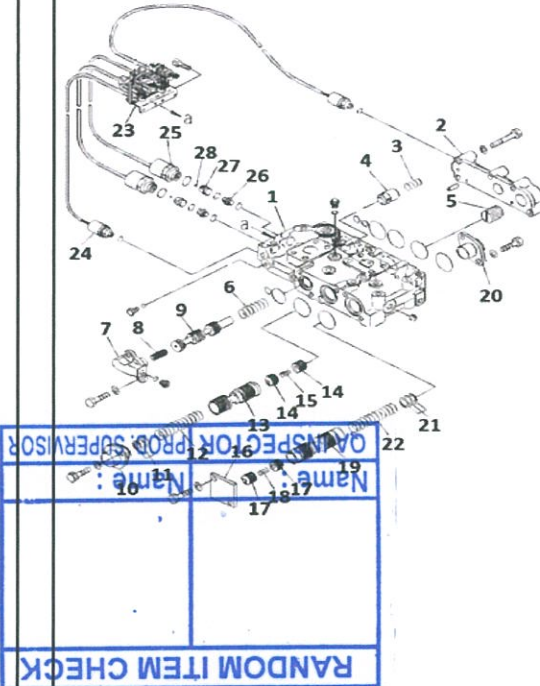
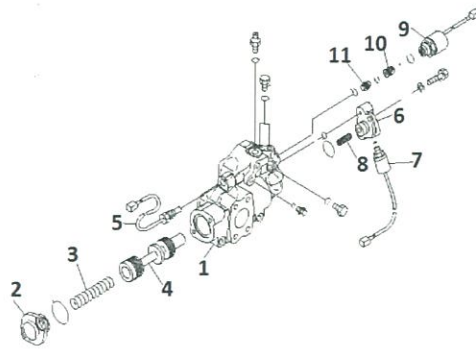
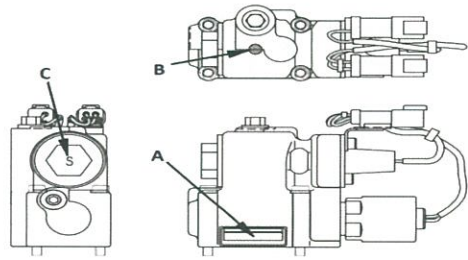
HD785-7

RO NO :

POWERTRAIN SECTION

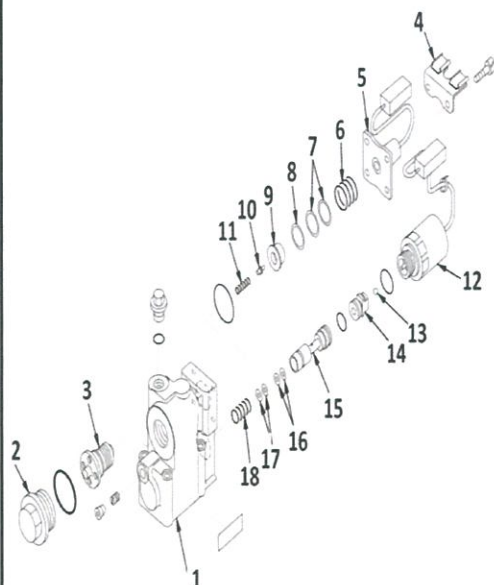
DISASSEMBLY CHECK SHEET CONTROL VALVE HD 785 - 7	WO NUMBER		DOCUMENT NO	
	COMP.MODEL		PUBLIC DATE	
	COMP.S/N		REVISION NO	R 00
	UNIT MODEL		PAGE	1 of 5

PROCESS	START		DISASSEMBLY BY	
	FINISH			

NO	PART TO REMOVE OR INSPECT	CONDITION TOBE INSPECTED	REUSE	SALVAGE	REPAIR	SKETCH DRAWING OR PHOTHOES	REMARK																				
	TORQUE CONVERTER ELIEF VALVE DIS ASSEMBLY	⚠ IMPORTANT ⚠ Check the quantity and thickness of the shims check the chipping ,scrath or banding of the spoll valve check the crack and abnormal worn of the valve body check the tension and banding of the spring - Remove pressure switch (24) and solenoid valve (25) from body (1) - Remove ball (28) valve (27) and plug (26) from valve body (1) - Remove cover (2) from body (1) - Remove piston (4) spring (3) (6) and piston (5) (14) spring (15) and main relief mvalve (13) from body (1) - Remove cover (7) and piston (8) from body (1) - remove selector valve (9) from body (1) - Remove vover (10) shims (11) and srping (12) - Remove cover (16) piston (17) spring (18) and torque conveter relief valve (19) from body (1) - Remove cover (20) shims (21) and spring (22) from body (1)																									
2	LUBRICATING RELIEF VALVE DISASSEMBLY	⚠ IMPORTANT ⚠ Examine valve body (1) and spool (4) as well as spring (3)for the existancecof plating film pieces and other metalic particles a When disassembling the valve, check that the spool moves smoothly - Remove all plug and temperature sensor (5) solenoid (9) and flow sensor (7) Remove valve (10)(11) and ball .from valve body (1) - Remove cover (2) and spring (3) from body (1) - remove cover (6) piston (8) and spool lubricating valve(4) frombody (1) .																									
3	ECMV DISASSEMBLY	⚠ IMPORTANT ⚠ there are 3 types of ECMV"s used which have the following distinction and differences <table><tr><th>Printing of name plate (A)</th><th>Color (B)</th><th>Stamp (C)</th><th>detect valve</th><th>Use clutch</th></tr><tr><td>D*****</td><td>—</td><td>—</td><td>Not slit</td><td>H,1st,2nd,3rd,R</td></tr><tr><td>K*****</td><td>—</td><td>—</td><td>Not equip</td><td>L/U</td></tr><tr><td>Q*****</td><td>Pink</td><td>S</td><td>Slit</td><td>L/4Th</td></tr></table>	Printing of name plate (A)	Color (B)	Stamp (C)	detect valve	Use clutch	D*****	—	—	Not slit	H,1st,2nd,3rd,R	K*****	—	—	Not equip	L/U	Q*****	Pink	S	Slit	L/4Th					
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DISASSEMBLY CHECK SHEET CONTROL VALVE HD 785 - 7	WO NUMBER		DOCUMENT NO	
	COMP.MODEL		PUBLIC DATE	
	COMP.S/N		REVISION NO	R 00
	UNIT MODEL		PAGE	2 of 5

PROCESS	START		DISASSEMBLY BY	
	FINISH			

NO	PART TO REMOVE OR INSPECT	CONDITION TOBE INSPECTED	REUSE	SALVAGE	REPAIR	SKETCH DRAWING OR PHOTHOES	REMARK
	ECMV DISASSEMBLY	⚠ IMPORTANT ⚠ Before removing the ECMV assembly from the transmission assembly, clean around it thoroughly so that dirt will not enter it or transmission.q Keep the removed parts so that they will not be damaged.Examine valve body (1) and spool (15) as well as spring (18)for the existancecof plating film pieces and other metallic particles Check the quantity and thickness of the shims check the chipping ,scrath or banding of the spoll valve check the crack and abnormal worn of the valve body check the tension and banding of the spring - Remove plug (2) and spool (3) from valve body (1).a When disassembling the valve, check that the spool moves smoothly - Remove bracket (4) and fill switch assembly(5). - Remove spring (6) and shims (7) and (8).a Check the quantity and thickness of the shims - Remove seat (9), spring (11) and pin (10). - Remove solenoid valve (12). Remove valve (14) and ball (13). - Remove spool (15), shims (16) and (17)and spring (18) Check the quantity and thickness of the shims					

NOTE

INSPECTION BY

MECHANIC

APPROVE BY

GROUP LEADER (GL)



PT SAPTAINDRA SEJATI

**CONTROL VALVE
MEASUREMENT
CHECKSHEET**

MACHINE MODEL

HD785-7

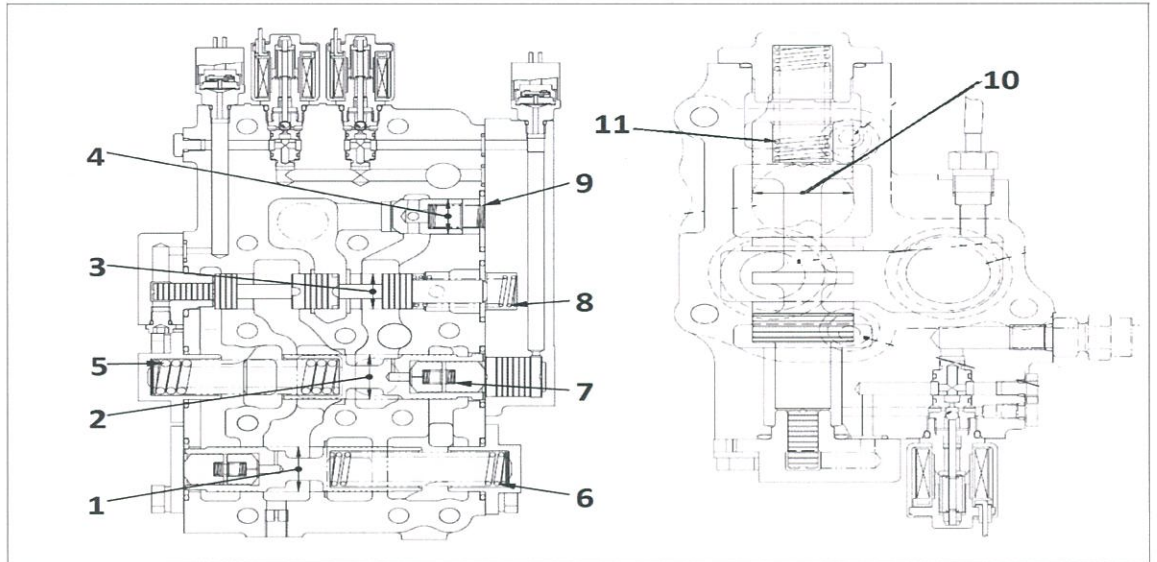
RO NO :

POWERTRAIN SECTION

MEASUREMENT CHECK SHEET CONTROL VALVE HD 785 - 7	WO NUMBER		DOCUMENT NO	
	COMP. MODEL		PUBLIC DATE	
	COMP.S/N		REVISION NO	R 00
	UNIT MODEL		PAGE	3 of 5

PROCESS	START FINISH		MEASUREMENT BY	
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NO	PART TO MEASURING OR INSPECT	STANDARD VALUE TABLE
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Unit: mm

No.	Check item	Criteria					Actual
		Standard size	Tolerance		Standard clearance	Clearance limit	
1	Clearance between the torque converter relief valve and valve body	32	-0.035 -0.045	+0.016 0	0.035 - 0.061	0.081	
2	Clearance between the main relief valve and valve body	32	-0.035 -0.045	+0.016 0	0.035 - 0.061	0.081	
3	Clearance between main flow rate selector valve and valve body	25	-0.035 -0.045	+0.013 0	0.035 - 0.058	0.078	
4	Clearance between check valve and valve body	25	-0.020 -0.041	+0.052 0	0.020 - 0.093	0.113	
5	Main relief valve spring	Standard size			Repair limit		
		Free length	Installed length	Installed load	Free length	Installed load	
		123	99.5	1213.2 N {123.8 kg}	119.3	1153.8 N {117.6 kg}	
6	Torque converter relief valve spring	137	93.5	331.2 N {33.8 kg}	132.9	315.0 N {32.1 kg}	
7	Poppet spring (Torque converter relief and main relief)	26	17.0	32.0 N {3.3 kg}	25.2	30.8 N {3.1 kg}	
8	Main flow rate selector valve spring	95	55.5	96.7 N {9.87 kg}	92.2	91.9 N {9.38 kg}	
9	Check valve spring	45	30	3.97 N {0.41 kg}	43.7	3.82 N {0.39 kg}	
10	Clearance between the spool and valve body	Standard size	Tolerance		Standard clearance	Clearance limit	
		40	-0.035 -0.045	+0.016 0	0.035 - 0.061	0.081	
11	Lubricating oil amount selector valve spring	Standard size			Repair limit		
		Free length	Installed length	Installed load	Free length	Installed load	
		103.5	65	134.3 N {13.7 kg}	100.4	127.4 N {13.0 kg}	

NOTE

MECHANIC

GROUP LEADER (GL)



PT SAPTAINDRA SEJATI

CONTROL VALVE

ASSEMBLY

CHECKSHEET

MACHINE MODEL

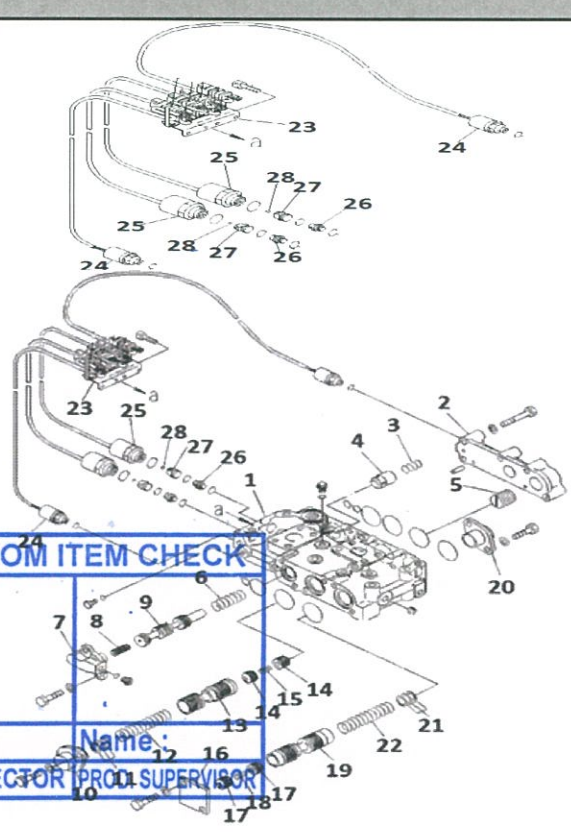
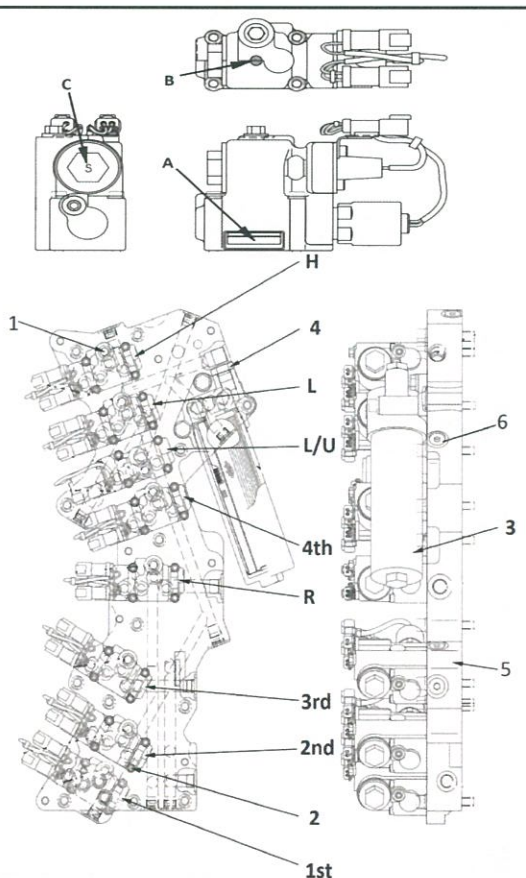
HD785-7

RO NO :

POWERTRAIN SECTION

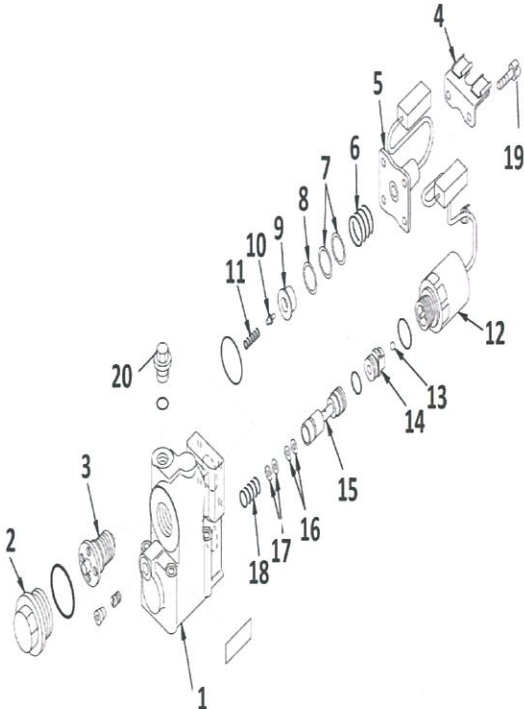
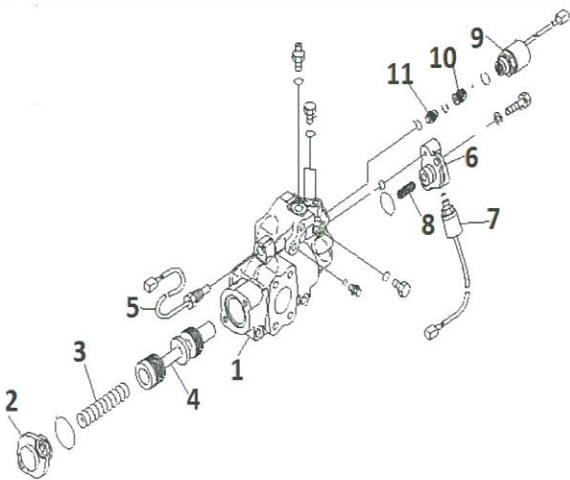
ASSEMBLY CHECK SHEET CONTROL VALVE HD 785 - 7	WO NUMBER		DOCUMENT NO	
	COMP.MODEL		PUBLIC DATE	
	COMP.S/N		REVISION NO	R 00
	UNIT MODEL		PAGE	4 of 5

PROCESS	START FINISH		ASSEMBLY BY	
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NO	PART TO INSTALL OR FIT	CONDITION TOBE ASSMBLY	MECH	GL/QA	SKETCH DRAWING OR PHOTHOES																																								
1	TORQUE CONVERTER RELIEF VALVE ASSEMBLY	⚠ IMPORTANT ⚠ Clean all parts,Inspect each part thoroughly and confirm that the part is free from dirt/dust, scratches, etc. coat the sliding surfaces of all parts with engine oil reassemble all spools and plungers into their original valve body bores. - install ball (28) valve (27) and plug (26) fit O-Ring to solenoid (25) Tightening valve (27) <table><tr><td>Standard</td><td>Actual</td></tr><tr><td>2.0 ± 0.2 Kgm</td><td></td></tr></table> - install piston valve (4) spring (3) piston valve (5) and torque Main relief relief valve (13) piston (14) spring (15) and Install valve body (2) to body (1) Tightening bolt cover <table><tr><td>Standard</td><td>Actual</td></tr><tr><td>6.0 ± 1.0 kgm</td><td></td></tr></table> - Install cover (20) shims (21) and spring (22) to body (1) - Install cover(16) piston(17) spring (18) and torque converter relief valve (19)to body (1) - Install vover (10) shims (11) and srping (12) - Install selector valve (9) to body (1) - Install cover (7) and piston (8) to body (1) - Install pressure switch (24) and solenoid valve (25) to body (1) Tightening solenoid <table><tr><td>Standard</td><td>Actual</td></tr><tr><td>5.0 ± 0.5 kgm</td><td></td></tr></table> RANDOM ITEM CHECK <table><tr><td>Name :</td><td>Name :</td></tr><tr><td>QA INSPECTOR</td><td>PRO SUPERVISOR</td></tr></table> 	Standard	Actual	2.0 ± 0.2 Kgm		Standard	Actual	6.0 ± 1.0 kgm		Standard	Actual	5.0 ± 0.5 kgm		Name :	Name :	QA INSPECTOR	PRO SUPERVISOR																											
Standard	Actual																																												
2.0 ± 0.2 Kgm																																													
Standard	Actual																																												
6.0 ± 1.0 kgm																																													
Standard	Actual																																												
5.0 ± 0.5 kgm																																													
Name :	Name :																																												
QA INSPECTOR	PRO SUPERVISOR																																												
2	ECMV SEAT ASSEMBLY	⚠ IMPORTANT ⚠ Inspect each part thoroughly and confirm that the part is free from dirt/dust, scratches. Be certain to reassemble all spools and plungers into their original valve body bores. there are 3 types of ECMV's used which have the following distinction and differences <table><tr><th>Printing of name plate (A)</th><th>Color (B)</th><th>Stamp (C)</th><th>detect valve</th><th>Use clutch</th></tr><tr><td>D*****</td><td>—</td><td>—</td><td>Not slit</td><td>H,1st,2nd,3rd,R</td></tr><tr><td>K*****</td><td>—</td><td>—</td><td>Not equip</td><td>L/U</td></tr><tr><td>Q*****</td><td>Pink</td><td>S</td><td>Slit</td><td>L/4Th</td></tr></table> Standard number of shims: 3 Standard thickness of shims: 1.5 mm (Thickness of 1 shim: 0.5 mm)Kinds of shims: 0.2, 0.5 mm Tightening bolt (2) <table><tr><td>Standard</td><td>Actual</td></tr><tr><td>0.9 ± 0.1 kgm</td><td></td></tr></table> Tightening plug (1) <table><tr><td>Standard</td><td>Actual</td></tr><tr><td>1.1 ± 0.1 kgm</td><td></td></tr></table> Tightening plug(4) <table><tr><td>Standard</td><td>Actual</td></tr><tr><td>1.8 ± 0.2 kgm</td><td></td></tr></table> Tightening filter(3) <table><tr><td>Standard</td><td>Actual</td></tr><tr><td>8.0 ± 0.5 kgm</td><td></td></tr></table> Tightening plug(6) <table><tr><td>Standard</td><td>Actual</td></tr><tr><td>2.7 ± 0.2 kgm</td><td></td></tr></table> 	Printing of name plate (A)	Color (B)	Stamp (C)	detect valve	Use clutch	D*****	—	—	Not slit	H,1st,2nd,3rd,R	K*****	—	—	Not equip	L/U	Q*****	Pink	S	Slit	L/4Th	Standard	Actual	0.9 ± 0.1 kgm		Standard	Actual	1.1 ± 0.1 kgm		Standard	Actual	1.8 ± 0.2 kgm		Standard	Actual	8.0 ± 0.5 kgm		Standard	Actual	2.7 ± 0.2 kgm				
Printing of name plate (A)	Color (B)	Stamp (C)	detect valve	Use clutch																																									
D*****	—	—	Not slit	H,1st,2nd,3rd,R																																									
K*****	—	—	Not equip	L/U																																									
Q*****	Pink	S	Slit	L/4Th																																									
Standard	Actual																																												
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2.7 ± 0.2 kgm																																													

ASSEMBLY CHECK SHEET		WO NUMBER	DOCUMENT NO
CONTROL VALVE		COMP.MODEL	PUBLIC DATE
HD 785 - 7		COMP.S/N	REVISION NO
		UNIT MODEL	PAGE
			R 00
			5 of 5

PROCESS	START	FINISH	ASSEMBLY BY

NO	PART TO INSTALL OR FIT	CONDITION TOBE ASSEMBLY	MECH	GL/QA	SKETCH DRAWING OR PHOTHOES																
3	ECMV VALVE ASSEMBLY	- Install shims (10) and (11) and spring (12) to spool (9).Install ball (13) and O-ring to valve (14) - Install spool (15) to valve body (1). - Install valve (14) to solenoid valve (12). Tightening valve (14) <table><tr><td>Standard</td><td>Actual</td></tr><tr><td>2.0 ± 0.2 Kgm</td><td></td></tr></table> - Fit the O-ring to solenoid valve (12) and install them to valve body (1). Tightening solenoid <table><tr><td>Standard</td><td>Actual</td></tr><tr><td>5.0 ± 0.5 kgm</td><td></td></tr></table> - Install pin (10) and spring (11) to seat (9). - Fit the O-ring to plug (2), install spool (3)to valve body (1), and install plug (2). Tightening plug (2) <table><tr><td>Standard</td><td>Actual</td></tr><tr><td>13.0 ± 1.5 kgm</td><td></td></tr></table> - Install seat (9), shims (8) and (7) and spring (6) to valve body (1). - Install switch assembly (5) and bracket (4) Tightening bolt (19) <table><tr><td>Standard</td><td>Actual</td></tr><tr><td>0.5 ± 0.2 kgm</td><td></td></tr></table>	Standard	Actual	2.0 ± 0.2 Kgm		Standard	Actual	5.0 ± 0.5 kgm		Standard	Actual	13.0 ± 1.5 kgm		Standard	Actual	0.5 ± 0.2 kgm				
Standard	Actual																				
2.0 ± 0.2 Kgm																					
Standard	Actual																				
5.0 ± 0.5 kgm																					
Standard	Actual																				
13.0 ± 1.5 kgm																					
Standard	Actual																				
0.5 ± 0.2 kgm																					
	LUBRICATING VALVE ASSEMBLY	▲ IMPORTANT ▲ Inspect each part thoroughly and confirm that the part is free from dirt/dust, scratches. Be certain to reassemble all spools and plungers into their original valve body bores. - Install all plug and temperature sensor (5) solenoid (9) and flow sensor (7) Install valve (10)(11) and ball .to valve body (1) - Install cover (2) and spring (3) to body (1) - Install cover (6) piston (8) and spool lubricating valve(4) to body (1) . Tightening bolt cover <table><tr><td>Standard</td><td>Actual</td></tr><tr><td>6.0 ± 1.0 kgm</td><td></td></tr></table>	Standard	Actual	6.0 ± 1.0 kgm																
Standard	Actual																				
6.0 ± 1.0 kgm																					

NOTE

INSPECTION BY

APPROVE BY

MECHANIC

GROUP LEADER (GL)



PT SAPTAINDRA SEJATI

TORQUE CONVERTER

ASSEMBLY

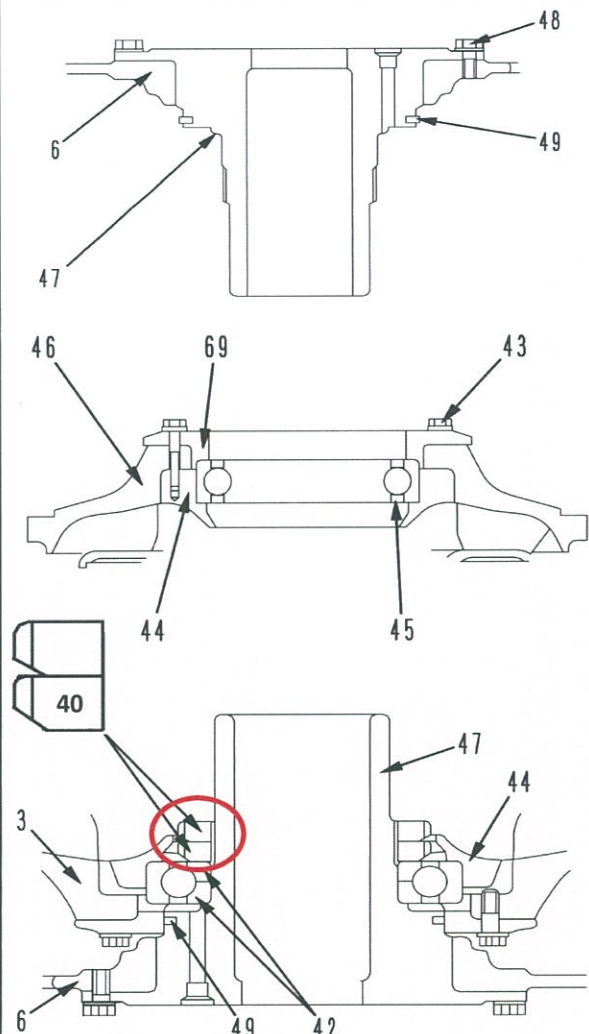
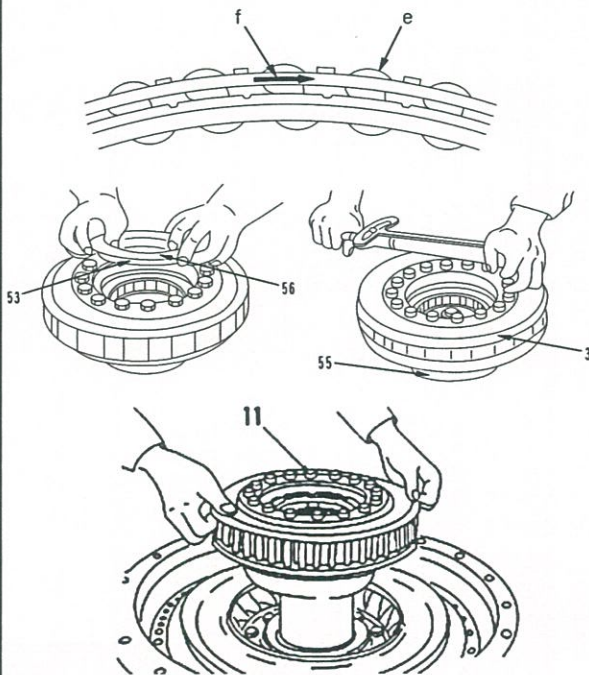
CHECKSHEET

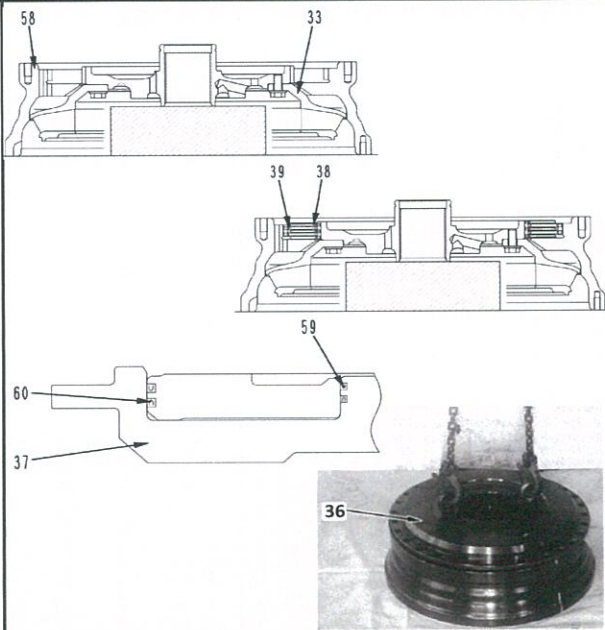
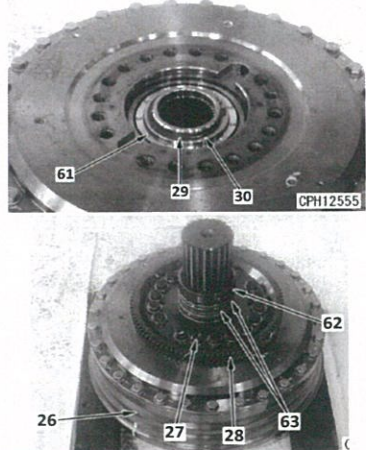
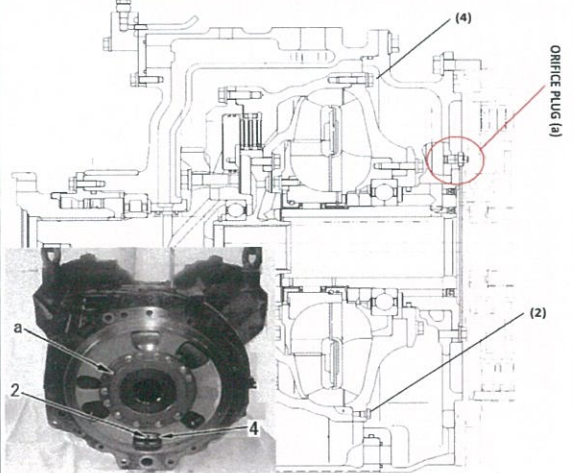
MACHINE MODEL

HD785-7

RO NO :

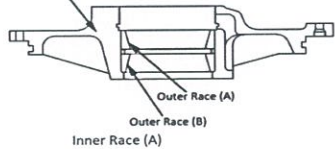
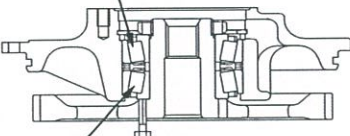
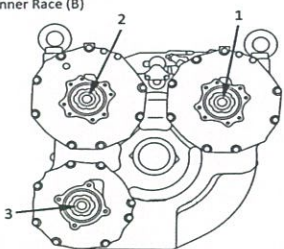
POWERTRAIN SECTION

ASSEMBLY CHECK SHEET		WO NUMBER	DOCUMENT NO	RC/QR04/PT/06/02																	
TORQUE CONVERTER		COMP.MODEL	PUBLIC DATE																		
HD 785 - 7		COMP.S/N	REVISION NO	REV 01																	
		UNIT MODEL	PAGE	1 OF 3																	
PROCESS	START		ASSEMBLY BY																		
	FINISH																				
NO	PART TO INSTALL OR ASSEMBLY	CONDITION TOBE INSTALL	MECH	GL/QA	SKETCH DRAWING OR PHOTHOES																
1	T/C SHAFT AND PUMP ASSEMBLY	- Install torque converter housing (6). - Install seal ring (49). - Don't ceel the wax's of seal ring (49) - Install shaft (47). Tightening Bolt <table border="1"> <tr> <th>Standard</th> <th>Actual</th> </tr> <tr> <td>11.5 ± 1 Kgm</td> <td></td> </tr> </table> - Reverse and install the shaft and torque converter housing assembly. Assemble the pump assembly in the following procedure: - Install bearing (45) to guide (44). - Install pump (46). - Install retainer (69) and tighten bolt Tightening Bolt <table border="1"> <tr> <th>Standard</th> <th>Actual</th> </tr> <tr> <td>6.25 ± 0.75 Kgm</td> <td></td> </tr> </table> - Install inner race (42) to shaft (47). - Install pump assembly (3). - Install remaining inner race (42). - Install nuts (40). 2 Nut threaded portion: Adhesive (LT-2) - Nut on the bearing side (first nut to install) Tightening Nut <table border="1"> <tr> <th>Standard</th> <th>Actual</th> </tr> <tr> <td>22.5 ± 2.5 Kgm</td> <td></td> </tr> </table> - Nut on the stator side (nut to install next) Tightening Nut <table border="1"> <tr> <th>Standard</th> <th>Actual</th> </tr> <tr> <td>47.5 ± 2.5 Kgm</td> <td></td> </tr> </table>	Standard	Actual	11.5 ± 1 Kgm		Standard	Actual	6.25 ± 0.75 Kgm		Standard	Actual	22.5 ± 2.5 Kgm		Standard	Actual	47.5 ± 2.5 Kgm				
Standard	Actual																				
11.5 ± 1 Kgm																					
Standard	Actual																				
6.25 ± 0.75 Kgm																					
Standard	Actual																				
22.5 ± 2.5 Kgm																					
Standard	Actual																				
47.5 ± 2.5 Kgm																					
2	Assembly of stator assembly	- Install one-way clutch to outer race (52). - a Install one-way clutch so that the "→" on its retainer will be directed clockwise. - a Take care not to damage sprag (e). - a Install the one-way clutch so that the arrow mark indicating the direction of racing will be clockwise (f) when seen from the direction of flange. Never remove the sprag. - Install ball (53) to bushing (56). - Chill bushing (56) in dry ice etc. - a Fix ball (53) with grease, match the position and install. - a Be careful not to damage the bushing contact surface. - Install Bolt Stator Adhesive (LT-2) Tightening Bolt <table border="1"> <tr> <th>Standard</th> <th>Actual</th> </tr> <tr> <td>6.75 ± 0.75 Kgm</td> <td></td> </tr> </table> Check that arrow showing the direction of free rotation point clockwise facing up Check that stator assembly can't rotate unclockwise.	Standard	Actual	6.75 ± 0.75 Kgm																
Standard	Actual																				
6.75 ± 0.75 Kgm																					

ASSEMBLY CHECK SHEET		WO NUMBER	DOCUMENT NO	RC/QR04/PT/06/02													
TORQUE CONVERTER		COMP.MODEL	PUBLIC DATE														
HD 785 - 7		COMP.S/N	REVISION NO	REV 01													
		UNIT MODEL	PAGE	2 OF 3													
PROCESS		START	ASSEMBLY BY														
		FINISH															
NO	PART TO INSTALL OR ASSEMBLY	CONDITION TOBE INSTALL	MECH	GL/QA	SKETCH DRAWING OR PHOTHOES												
3	TURBIN & PISTON ASSEMBLY	- Install inner gear to turbine and tighten bolts  Adhesive (LT -2)  Tightening Bolt <table border="1"> <tr> <th>Standard</th> <th>Actual</th> </tr> <tr> <td>11.25 ± 1.25 Kgm</td> <td></td> </tr> </table> - Set case (58) and turbine assembly (33).a Place blocks under the turbine to align the disc installation surfaces. - Install disc (38) and plate (39). - Install bearing (68) to housing. - Install seal rings (59) and (60) to piston (37). - Install piston (37) to housing (36). - Install housing (36)  Adhesive (LT -2)  Tightening Bolt <table border="1"> <tr> <th>Standard</th> <th>Actual</th> </tr> <tr> <td>11.25 ± 1.25 Kgm</td> <td></td> </tr> </table>	Standard	Actual	11.25 ± 1.25 Kgm		Standard	Actual	11.25 ± 1.25 Kgm								
Standard	Actual																
11.25 ± 1.25 Kgm																	
Standard	Actual																
11.25 ± 1.25 Kgm																	
4	BEARING & DRIVE GEAR INSTALLATION	- Press-fit bearing (61). - a Press-fit the inner race and outer race at the same time. - Install spacer (30) and install snap ring (29). - Use the press-fitting tool to press-fit inner race (62) to shaft (20). - Install seal ring (63). - Install drive gear assembly (27) to turbine clutch assembly (26) and tighten the bolts.  Adhesive (LT -2)  Tightening Bolt <table border="1"> <tr> <th>Standard</th> <th>Actual</th> </tr> <tr> <td>28.25 ± 3.25 Kgm</td> <td></td> </tr> </table>	Standard	Actual	28.25 ± 3.25 Kgm												
Standard	Actual																
28.25 ± 3.25 Kgm																	
5	TURBIN & PUMP INSTALLATION	- Tighten pump mounting bolt (4) and drain plug (2)  Adhesive (LT -2)  Tightening Bolt <table border="1"> <tr> <th>Standard</th> <th>Actual</th> </tr> <tr> <td>5.5 ± 0.5 Kgm</td> <td></td> </tr> </table>  Adhesive (LT -2)  Tightening Plug <table border="1"> <tr> <th>Standard</th> <th>Actual</th> </tr> <tr> <td>1.15 ± 0.15 Kgm</td> <td></td> </tr> </table> - Install orifice plug (a)  Adhesive (LT -2)  Tightening Plug <table border="1"> <tr> <th>Standard</th> <th>Actual</th> </tr> <tr> <td>1.15 ± 0.15 Kgm</td> <td></td> </tr> </table>	Standard	Actual	5.5 ± 0.5 Kgm		Standard	Actual	1.15 ± 0.15 Kgm		Standard	Actual	1.15 ± 0.15 Kgm				
Standard	Actual																
5.5 ± 0.5 Kgm																	
Standard	Actual																
1.15 ± 0.15 Kgm																	
Standard	Actual																
1.15 ± 0.15 Kgm																	

ASSEMBLY CHECK SHEET TORQUE CONVERTER HD 785 - 7	WO NUMBER		DOCUMENT NO	RC/QR04/PT/06/02
	COMP.MODEL		PUBLIC DATE	
	COMP.S/N		REVISION NO	REV 01
	UNIT MODEL		PAGE	3 OF 3

PROCESS	START FINISH		ASSEMBLY BY		
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NO	PART TO INSTALL OR ASSEMBLY	CONDITION TOBE INSTALL	MECH	GL/QA	SKETCH DRAWING OR PHOTHOES																		
6	PTO ASSEMBLY	Check manufacturing no of inner and outer race - and match marks (A and A, B and B) with press methode <table border="1"><tr><td>1</td><td></td></tr><tr><td>2</td><td></td></tr><tr><td>3</td><td></td></tr></table> - Coat with TO10 or TO30 on bearing - Turn 10 times and recheck torque  Engine oil SAE 30  Tightening Nut <table border="1"><thead><tr><th>Standard</th><th>Actual</th></tr></thead><tbody><tr><td>36 ± 4 Kgm</td><td>1</td></tr><tr><td></td><td>2</td></tr><tr><td></td><td>3</td></tr></tbody></table> - Install PTO Assembly  Tightening Bolt <table border="1"><thead><tr><th>Standard</th><th>Actual</th></tr></thead><tbody><tr><td>6.25 ± 0.75 Kgm</td><td></td></tr></tbody></table>	1		2		3		Standard	Actual	36 ± 4 Kgm	1		2		3	Standard	Actual	6.25 ± 0.75 Kgm				PTO HOUSING  Outer Race (A) Outer Race (B) Inner Race (A)  Inner Race (B)  1 2 3
1																							
2																							
3																							
Standard	Actual																						
36 ± 4 Kgm	1																						
	2																						
	3																						
Standard	Actual																						
6.25 ± 0.75 Kgm																							

INSPECTION BY

MECHANIC

APPROVE BY

GROUP LEADER (GL)



PT SAPTAINDRA SEJATI

TRANSMISSION

ASSEMBLY

CHECKSHEET

MACHINE MODEL

HD785-7

RO NO :

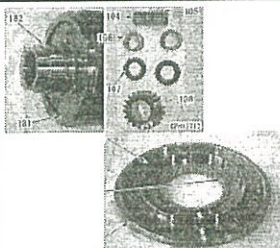
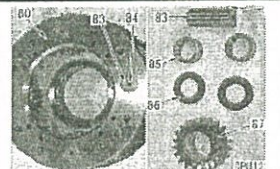
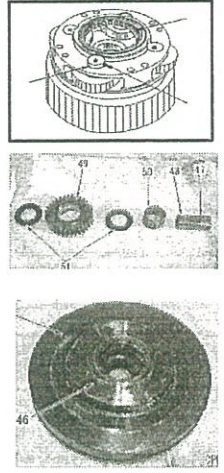
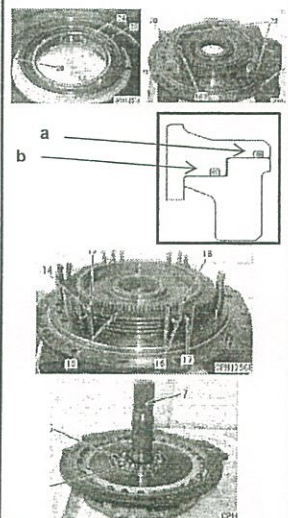
POWERTRAIN SECTION



PT SAPTAINDRA SEJATI
NAROGONG REBUILD CENTER

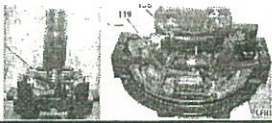
ASSEMBLY CHECK SHEET TRANSMISSION (HD785-7)	WO NO.		DOCUMENT NO.	
	COMP. MODEL	Torque Flow T/M	PUBLIC DATE	15/10/2010
	COMP. S/N		REVISION NO.	00
	UNIT MODEL	HD785-7	PAGE	1 of 4

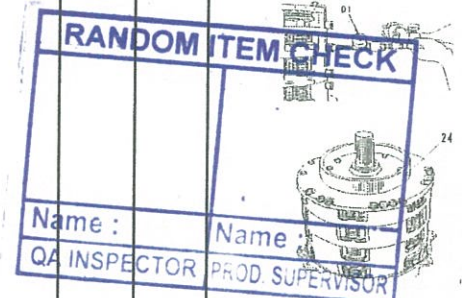
PROCESS	START	ASSEMBLY BY
	FINISH	

NO.	PARTS	CHECK POINT	SPECIFICATION	SIGN			REMARKS (SKETCH, PHOTOS, COMMENTS)
				MEC	SPV	QA	
1	ASSEMBLY OF NO 5,6 CARRIER ASS'Y	- Clean all parts, check for dirt or damage, and correct any burrs - Check that gear rotates smoothly - Install cover to carrier assy - Torque monting bolt Act : Kgm	Push shaft with plastic hammer / copper hammer Std torque bolt 6 - 7.5 kgm				
2	ASSEMBLY OF NO 4, 5 CARRIER ASS'Y	- Clean all parts, check for dirt or damage, and correct any burrs - Check that gear rotates smoothly - Install cover to carrier assy	push shaft with plastic hammer/ copper hammer				
3	ASSEMBLY OF CARRIER AND CLUTCH LOW AND CLUTCH 4	- Check that gear rotates smoothly - Torque Bolt Plate Act: Kgm - Check that teclon with the groove on side taking pressure - Torque bolt cilynder Act: Kgm - Torque bolt housing Act: Kgm	Std torque bolt 10 - 12.5 kgm LT - 2 Std torque bolt 6 - 7.5 kgm LT - 2 Std torque bolt 10 - 12.5 kgm LT - 2				
4	ASSEMBLY OF HIGH CLUTCH	- Install bearing to cage and fix with snap ring press fit - Install cage and ring gear - Torque bolt Act : kgm Install disc,plate and piston plate - set the seal ring (a) ,(b) groove portion to the pressure receiving side - Install clutch housing torque bolt Act : kgm	Std torque bolt 16 - 20 kgm LT - 2 Std torque bolt 16 - 20 kgm LT - 2				



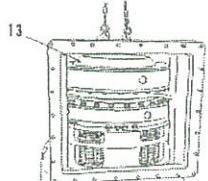
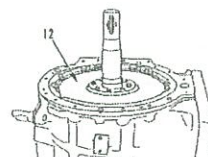
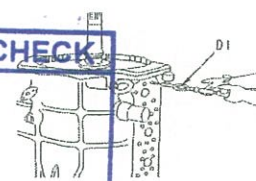
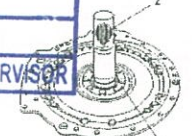
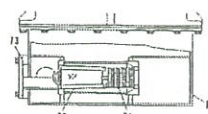
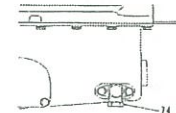
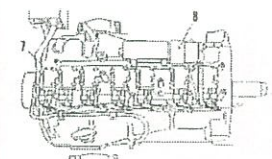
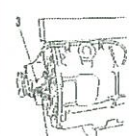
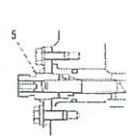
ASSEMBLY CHECK SHEET TRANSMISSION (HD785-7)	WO NO.	DOCUMENT NO.	
	COMP. MODEL	Torque Flow T/M	PUBLIC. DATE
	COMP. S/N		REVISION NO.
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			15/10/2010
			00
			2 of 4

NO.	PARTS	CHECK POINT	SPECIFICATION	SIGN			REMARKS (SKETCH, PHOTOS, COMMENTS)																																													
				MEC	SPV	QA																																														
5	ASSEMBLY OF TRANSMISSION ASS'Y	- Make sure that oil seal lip not twist when assembly - Install oil seal to cage - install bearing by the inner race side to install cage torque monting bolt Act. : kgm	Lip surface of oil seal G2-L1 Std torque bolt 25 - 31 kgm LT - 2				 																																													
		- install dowel pin to carrier and then - install collar - install ring gear to cover assy - install guide pin - install spring, disc, plate and piston plate Free Length Of Spring - Push tie bolt by hand and don't use impact wrench until the bolt has mated with the tap thread Torque tie bolt Act: Kgm - Check piston stroke Actual Air pressure Act: Kg/cm² Actual Piston Stroke<table><tr><th>No of Clutch</th><th>Act</th></tr><tr><td>1st</td><td></td></tr><tr><td>2nd</td><td></td></tr><tr><td>R</td><td></td></tr><tr><td>3rd</td><td></td></tr></table> Torque boltring gear Act: Kgm - Check Seal ring groove on the side taking the pressure Torque bolt cover Act: Kgm	No of Clutch	Act	1st		2nd		R		3rd		<table><tr><th>No of Clutch</th><th colspan="2">Free Length of spring</th></tr><tr><td></td><th>HD785-7</th><th>HD465-7</th></tr><tr><td>1st</td><td>68.0</td><td>68.0</td></tr><tr><td>2nd</td><td>68.0</td><td>68.0</td></tr><tr><td>R</td><td>84.0</td><td>91</td></tr><tr><td>3rd</td><td>41.5</td><td>52.0</td></tr><tr><td>H</td><td>52.0</td><td>42.0</td></tr></table> Std. torque Kgm <table><tr><td>HD785-5</td><td>25.0 - 28.0</td></tr></table> Std. Air pressure 3~6 Kg/cm² Std. Piston Stroke <table><tr><th>No of Clutch</th><th colspan="2">HD785-7</th></tr><tr><td>1st</td><td>6.8</td><td></td></tr><tr><td>2nd</td><td>6.3</td><td></td></tr><tr><td>R</td><td>6.0</td><td></td></tr><tr><td>3rd</td><td>4.0</td><td></td></tr></table> Use thread tightener LT2 Std. Torque 18 ± 2 Kgm Use thread tightener LT2 Std. Torque 6.7 ± 0.7 Kgm	No of Clutch	Free Length of spring			HD785-7	HD465-7	1st	68.0	68.0	2nd	68.0	68.0	R	84.0	91	3rd	41.5	52.0	H	52.0	42.0	HD785-5	25.0 - 28.0	No of Clutch	HD785-7		1st	6.8		2nd	6.3		R	6.0		3rd	4.0		
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ASSEMBLY CHECK SHEET TRANSMISSION (HD785-7)	WO NO.		DOCUMENT NO.	
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NO.	PARTS	CHECK POINT	SPECIFICATION	SIGN			REMARKS (SKETCH, PHOTOS, COMMENTS)
				MEC	SPV	QA	
		- Torque bolt housing Act: Kgm - Check that 4th clutch rotates smoothly after install transmission case - Torque bolt Act: Kgm - Check low clutch and high clutch rotates smoothly after install - Check piston stroke or movement of high clutch - Torque bolt cover high clutch Act: Kgm - Torque bolt plat oil pan Act: Kgm - Torque bolt oil pan Act: Kgm - Torque bolt cover strainer and drain plug Act: Kgm - Torque bolt valve seat Act: Kgm - Torque ECMV valve Act: Kgm - Torque bolt lubricating valve Act: Kgm - Torque bolt speed sensor and techograph sensor Act: Kgm - Check that connector and wiring harness is correctly and good condition	Bearing heat 100°C~110°C Use thread tightener LT2 Std. Torque 11.5± 1 Kgm Use thread tightener LT2 Std. Torque 6.7± 0.7 Kgm Use guide shaft when install transmission case Std. Torque 6.7± 0.7 Kgm LT2 Std. Torque 6.7± 0.7 Kgm LG6 Std. Torque 6.7± 0.7 Kgm Std. Torque 6.7± 0.7 Kgm Std. Torque 3.2± 0.3 Kgm ECMV low clutch refer and follow PSN : AT00089 Std. Torque 6.7± 0.7 Kgm Std. Torque 3.2± 0.3 Kgm				         

RANDOM ITEM CHECK

Name :

Name :

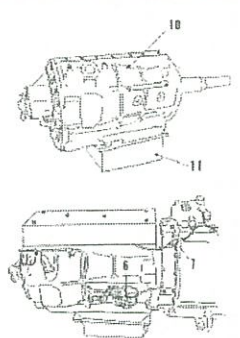
QA INSPECTOR

PROD. SUPERVISOR



PT SAPTAINDRA SEJATI
NAROGONG REBUILD CENTER

ASSEMBLY CHECK SHEET TRANSMISSION (HD785-7)	WO NO.		DOCUMENT NO.	
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NO.	PARTS	CHECK POINT	SPECIFICATION	SIGN			REMARKS (SKETCH, PHOTOS, COMMENTS)
				MEC	SPV	QA	
		- Torque bolt transmission case Act: Kgm - Torque bolt connection T/M to torque converter Act: Kgm	Std. Torque 28.5± 3 Kgm Std. Torque 28.5± 3 Kgm				

NOTES

ASSEMBLY BY

INSPECTED BY

APPROVED BY

(MECHANIC)

(QA)

(SUPERVISOR)